Scott's Factorial Series

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Abstract

This paper studies the function defined by the series

$$S(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{e^n (n!)^2} \left[\sin\left(\frac{x}{(n+1)!}\right)\right]^{2n+1}.$$

The series combines alternating signs, factorial coefficients, and increasing odd powers of the sine function. We prove that it converges absolutely and uniformly on the real line. By extending the analysis to complex inputs, we show that the resulting function is entire and therefore has derivatives of every order. We also derive its Taylor expansion near the origin, establish global bounds, and obtain an error estimate for finite approximations. Finally, we examine the periodicity of the partial sums and discuss possible generalizations.

Keywords: factorial series; uniform convergence; entire functions; Taylor expansion; approximation error.

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1 Introduction

Infinite series provide a useful way to define functions and study their properties. The behavior of a series often depends on the relationship between its coefficients and its individual terms.

Factorial coefficients are particularly useful because they can force rapid convergence even when the remaining factors vary with the input.

Consider the function

$$S(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{e^n (n!)^2} \left[\sin\left(\frac{x}{(n+1)!} \right) \right]^{2n+1}.$$

Three features distinguish this series. Its coefficients decrease factorially, the argument of the sine contains a factorial, and the exponent of the sine increases with the summation index. The signs also alternate.

These features raise several questions. Does the series converge for every real input? Is its sum continuous and differentiable? Can the function be represented by a Taylor series, and how accurately can it be approximated by a finite sum?

The first part of this paper establishes convergence and continuity. The subsequent sections examine stronger regularity properties, behavior near the origin, global bounds, and numerical approximation. The arguments use standard results on uniform convergence and holomorphic functions; see, for example, Rudin [2] and Conway [1].

2 Definition and Basic Properties

2.1 The series and its partial sums

Definition 2.1 (Scottlevoy's Factorial Series)**.** For $x \in \mathbb{R}$, define

$$S(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{e^n (n!)^2} \left[\sin\left(\frac{x}{(n+1)!} \right) \right]^{2n+1}.$$

Here, e is Euler's number and $n!$ denotes the factorial of the nonnegative integer n, with $0! = 1$.

Write the n-th term as

$$a_n(x) = \frac{(-1)^n}{e^n (n!)^2} \left[\sin\left(\frac{x}{(n+1)!} \right) \right]^{2n+1}.$$

The N-th partial sum is

$$S_N(x) = \sum_{n=0}^{N} a_n(x). \tag{1}$$

Each partial sum is a finite combination of trigonometric functions. The infinite series is obtained by taking the limit of these partial sums, provided that the limit exists.

2.2 The first terms

The first five terms are displayed in Table 1.

Table 1: The first five terms of the series.

n	Sign	Coefficient	Sine factor
0	$+$	1	$\sin(x)$
1	$-$	$\frac{1}{e}$	$\sin^3\left(\frac{x}{2}\right)$
2	$+$	$\frac{1}{4e^2}$	$\sin^5\left(\frac{x}{6}\right)$
3	$-$	$\frac{1}{36e^3}$	$\sin^7\left(\frac{x}{24}\right)$
4	$+$	$\frac{1}{576e^4}$	$\sin^9\left(\frac{x}{120}\right)$

The factorials enter the series in two different ways. The coefficient decreases as n increases, while the argument of the sine becomes smaller for any fixed x. The increasing exponent further changes the contribution of each term.

These effects suggest that the later terms should become small. A convergence proof is needed, however, to establish that their infinite sum is well-defined.

3 Convergence and Continuity

3.1 Absolute and uniform convergence

Theorem 3.1. *The series defining $S(x)$ converges absolutely and uniformly on $\mathbb{R}$.*

Proof. For every real x,

$$\left|\sin\left(\frac{x}{(n+1)!}\right)\right| \leq 1.$$

It follows that

$$|a_n(x)| \leq \frac{1}{e^n (n!)^2}.$$

Consider the numerical series

$$\sum_{n=0}^{\infty} \frac{1}{e^n (n!)^2}.$$

The ratio of consecutive terms is

$$\frac{1/[e^{n+1}((n+1)!)^2]}{1/[e^n (n!)^2]} = \frac{1}{e(n+1)^2}.$$

This ratio tends to zero as $n \to \infty$, so the numerical series converges by the ratio test.

The Weierstrass M-test therefore proves that the original series converges uniformly on $\mathbb{R}$ [2]. Since the same bound applies to the absolute values of its terms, the convergence is also absolute at every real x. □

Corollary 3.2. *The function S is continuous on $\mathbb{R}$.*

Proof. Each $a_n(x)$ is continuous, so every partial sum $S_N(x)$ is continuous. The uniform convergence of S_N to S then implies that S is continuous on $\mathbb{R}$. □

3.2 A global bound

The same estimate used in the convergence proof also gives a bound on the value of the function.

Proposition 3.3. *For every $x \in \mathbb{R}$,*

$$|S(x)| \leq \sum_{n=0}^{\infty} \frac{1}{e^n (n!)^2} \leq e^{1/e}.$$

Proof. By the triangle inequality and the bound on each term,

$$|S(x)| \leq \sum_{n=0}^{\infty} |a_n(x)| \leq \sum_{n=0}^{\infty} \frac{1}{e^n (n!)^2}.$$

Since $n! \geq 1$, we have

$$\frac{1}{(n!)^2} \leq \frac{1}{n!}.$$

Consequently,

$$\sum_{n=0}^{\infty} \frac{1}{e^n (n!)^2} \leq \sum_{n=0}^{\infty} \frac{(1/e)^n}{n!} = e^{1/e}.$$

This proves the result. □

Thus, although the sine factors depend on x, the function remains bounded on the entire real line.

4 Differentiability and Analyticity

4.1 The first derivative

Differentiating an individual term gives

$$a_n'(x) = \frac{(-1)^n (2n+1)}{e^n (n!)^2 (n+1)!} \times \sin^{2n}\left(\frac{x}{(n+1)!}\right) \cos\left(\frac{x}{(n+1)!}\right). \tag{2}$$

Because the sine and cosine are bounded in absolute value by 1 on the real line,

$$|a_n'(x)| \leq \frac{2n+1}{e^n (n!)^2 (n+1)!}.$$

The numerical series of these bounds converges by the ratio test. Therefore, the derivative series converges uniformly on $\mathbb{R}$. The theorem on differentiating uniformly convergent series implies that S is continuously differentiable and that

$$S'(x) = \sum_{n=0}^{\infty} a_n'(x).$$

This establishes differentiability everywhere, rather than leaving it as an open question.

4.2 Higher derivatives

The regularity of S can be strengthened further by considering complex inputs.

Theorem 4.1. *The series defining S converges locally uniformly on $\mathbb{C}$. Its sum therefore extends to an entire function, and all derivatives of S exist and are continuous on $\mathbb{R}$.*

Proof. Extend each term to a complex variable z:

$$a_n(z) = \frac{(-1)^n}{e^n(n!)^2} \sin^{2n+1}\left(\frac{z}{(n+1)!}\right).$$

Each a_n is entire.

For every complex number w,

$$|\sin w| \leq e^{|w|}.$$

Fix $R > 0$ and suppose that $|z| \leq R$. Then

$$|a_n(z)| \leq \frac{1}{e^n(n!)^2} \exp\left(\frac{(2n+1)R}{(n+1)!}\right).$$

The sequence

$$\frac{2n+1}{(n+1)!}$$

is bounded above by $3/2$ for all nonnegative integers n. Hence

$$|a_n(z)| \leq \frac{e^{3R/2}}{e^n(n!)^2}.$$

The right-hand side is summable and independent of z within the disk $|z| \leq R$. The Weierstrass M-test therefore gives uniform convergence on every closed disk.

A locally uniform limit of entire functions is entire [1]. Consequently, the complex extension of S is entire, which proves the result. □

In particular, S is real analytic: near every real point, it is represented by a convergent Taylor series. The Taylor series about the origin in fact converges for every complex input.

5 Behavior Near the Origin

5.1 The value and derivative at zero

Since every term contains a positive power of a sine factor,

$$S(0) = 0.$$

The first term of the series is $\sin x$. Every remaining term vanishes to order at least three at the origin. It follows that

$$S'(0) = 1.$$

The function therefore behaves like the identity function near zero, in the sense that

$$S(x) = x + O(x^3) \qquad \text{as } x \to 0.$$

In particular,

$$\lim_{x \to 0} \frac{S(x)}{x} = 1.$$

This also shows that $S(x)$ is positive for sufficiently small positive x and negative for sufficiently small negative x.

5.2 Taylor expansion

The Taylor expansion of the sine function is

$$\sin t = t - \frac{t^3}{6} + \frac{t^5}{120} + O(t^7).$$

For the second term of S, we also use

$$\sin^3 t = t^3 - \frac{t^5}{2} + O(t^7).$$

The terms with indices $n = 0, 1, 2$ contribute to the Taylor expansion through degree five. Every term with $n \geq 3$ vanishes to order at least seven at zero. Combining these contributions

gives

$$S(x) = x - \left(\frac{1}{6} + \frac{1}{8e}\right) x^3 + \left(\frac{1}{120} + \frac{1}{64e} + \frac{1}{31104e^2}\right) x^5 + O(x^7). \tag{3}$$

The coefficient of x^3, for example, receives contributions from both $\sin x$ and $-e^{-1}\sin^3(x/2)$. The coefficient of x^5 receives contributions from the first three terms.

More generally, the term $a_n(x)$ vanishes to order $2n+1$ at zero. Therefore, the coefficient of x^{2k+1} in the Taylor expansion receives contributions only from indices $n \le k$. Each Taylor coefficient is thus determined by a finite calculation, even though S is defined by an infinite series.

6 Symmetry and Periodicity

6.1 Odd symmetry

The exponent $2n+1$ is odd, and the sine function is odd. Thus

$$a_n(-x) = -a_n(x)$$

for every n and every real x.

Summing the series gives

$$S(-x) = -S(x).$$

Hence S is an odd function. In particular, every even derivative of S at the origin vanishes:

$$S^{(2k)}(0) = 0, \qquad k = 0, 1, 2, \ldots.$$

This agrees with the Taylor expansion in Equation (3), which contains only odd powers.

6.2 Periodicity of the partial sums

Although the full series is defined on the entire real line, each finite partial sum is periodic.

Proposition 6.1. *For every nonnegative integer N, the partial sum S_N has period $2\pi(N+1)!$.*

Proof. For $0 \leq n \leq N$, replacing x by $x + 2\pi(N+1)!$ changes the argument of the sine by

$$\frac{2\pi(N+1)!}{(n+1)!}.$$

Because $(n+1)!$ divides $(N+1)!$, this quantity is an integer multiple of 2π.

The sine factor in each term is therefore unchanged. Every term of S_N has the stated period, so their sum has that period as well. □

The periods supplied by this proposition increase with N. The proposition establishes periodicity of every finite approximation, but it does not establish that the infinite sum has a finite period.

7 Numerical Approximation

7.1 A uniform error estimate

The partial sums provide natural approximations to $S(x)$. Their accuracy can be bounded independently of the value of x.

Theorem 7.1. *For every nonnegative integer N and every $x \in \mathbb{R}$,*

$$|S(x) - S_N(x)| \leq \sum_{n=N+1}^{\infty} \frac{1}{e^n (n!)^2}.$$

Moreover,

$$|S(x) - S_N(x)| \leq \frac{1}{e^{N+1}((N+1)!)^2} \frac{1}{1 - \dfrac{1}{e(N+2)^2}}.$$

Proof. The first inequality follows directly from the triangle inequality and the bound

$$|a_n(x)| \leq \frac{1}{e^n (n!)^2}.$$

Let

$$b_n = \frac{1}{e^n (n!)^2}.$$

For $n \geq N+1$,

$$\frac{b_{n+1}}{b_n} = \frac{1}{e(n+1)^2} \leq \frac{1}{e(N+2)^2}.$$

Set

$$r_N = \frac{1}{e(N+2)^2}.$$

Since $0 < r_N < 1$, comparison with a geometric series gives

$$\sum_{n=N+1}^{\infty} b_n \leq b_{N+1} \sum_{k=0}^{\infty} r_N^k = \frac{b_{N+1}}{1-r_N}.$$

Substitution of b_{N+1} proves the second inequality. □

This estimate gives a rigorous stopping criterion for numerical calculations. Once the upper bound is smaller than a chosen tolerance, the partial sum approximates $S(x)$ to within that tolerance for every real x.

The bound is conservative because it replaces every sine factor by its maximum possible absolute value. For a particular input, the actual error may be substantially smaller.

8 Possible Generalizations

The structure of the series suggests several variations. For example, one may introduce a parameter $\lambda > 0$ and consider

$$S_\lambda(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{\lambda^n (n!)^2} \left[\sin\left(\frac{x}{(n+1)!}\right)\right]^{2n+1}.$$

For each fixed $\lambda > 0$, the same comparison argument proves absolute and uniform convergence on $\mathbb{R}$. Indeed,

$$|a_{n,\lambda}(x)| \leq \frac{1}{\lambda^n (n!)^2},$$

and the ratio of consecutive terms in the majorant is

$$\frac{1}{\lambda(n+1)^2} \longrightarrow 0.$$

Thus, the basic convergence result does not depend on the particular choice $\lambda = e$. Other positive parameters can be studied using the same method.

A second variation replaces the factorial in the sine argument by a more general sequence:

$$T(x) = \sum_{n=0}^{\infty} c_n \left[\sin\left(\frac{x}{q_n}\right)\right]^{m_n},$$

where c_n, q_n, and m_n are chosen sequences. If

$$\sum_{n=0}^{\infty} |c_n| < \infty,$$

then $|\sin t| \leq 1$ immediately gives absolute and uniform convergence on the real line, provided the remaining terms have the stated form.

This condition is sufficient rather than necessary. If the coefficients are not absolutely summable, the decay of the sine factors may still ensure convergence, but a different argument would be needed.

These generalizations show that the convergence proof belongs to a broader class of series. A more detailed study could examine which choices of coefficients, arguments, and exponents lead to stronger properties or more useful approximations.

9 Conclusion

Scottlevoy's Factorial Series defines a bounded, odd, real-analytic function on the real line. The Weierstrass M-test establishes absolute and uniform convergence, and an extension to complex inputs shows that the function is entire. In particular, it has continuous derivatives of every order.

The Taylor expansion describes the behavior near the origin, where $S(0) = 0$ and $S'(0) = 1$. A global bound controls the size of the function, while an explicit remainder estimate controls the error of its finite partial sums. The partial sums are also periodic, although their periods increase with the truncation index.

Several questions remain open within the scope of this study. These include determining whether the infinite sum has a finite period, finding sharper bounds for its derivatives, and investigating how the function changes under different coefficient sequences. These questions provide ways to extend the analysis beyond the basic convergence and regularity results established here.

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References

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